

[S] 对照 hsize=T (应同旧 minipage 断行)

已知正方体

$ABCD-A_1B_1C_1D_1$ 中, $\overrightarrow{A_1E} = \frac{1}{4}$

$\overrightarrow{A_1C_1}$, 若

$\overrightarrow{AE} = x\overrightarrow{AA_1} + y(\overrightarrow{AB} + \overrightarrow{AD})$, 则

$x = \underline{\hspace{2cm}}$, $y = \underline{\hspace{2cm}}$.

[T] hangindent=-36.432mm hangafter=-5 (窄 5 行 + 通栏)

已知

正方体 $ABCD-A_1B_1C_1D_1$ 中,

$\overrightarrow{A_1E} = \frac{1}{4} \overrightarrow{A_1C_1}$,

若 $\overrightarrow{AE} = x\overrightarrow{AA_1} + y(\overrightarrow{AB} + \overrightarrow{AD})$,

则 $x = \underline{\hspace{2cm}}$, $y = \underline{\hspace{2cm}}$.

[U] hangindent=-36.432mm hangafter=-4 (长文本, 窄 4 行 + 通栏)

[详解] 过点 B, D 分别向 AC

作垂线, 垂足分别为 M, N ,

则可得 $AM = \frac{1}{2}$, $BM = \frac{\sqrt{3}}{2}$,

$CN = \frac{1}{2}$, $DN = \frac{\sqrt{3}}{2}$,

$MN = 1$. 由于 $\overrightarrow{BD} = \overrightarrow{BM} + \overrightarrow{MN} + \overrightarrow{ND}$, 所以

$|\overrightarrow{BD}|^2 = (\overrightarrow{BM} + \overrightarrow{MN} + \overrightarrow{ND})^2$

$= |\overrightarrow{BM}|^2 + |\overrightarrow{MN}|^2 + |\overrightarrow{ND}|^2 + 2(\overrightarrow{BM} \cdot \overrightarrow{MN} + \overrightarrow{MN} \cdot \overrightarrow{ND} + \overrightarrow{BM} \cdot \overrightarrow{ND})$

$= (\frac{\sqrt{3}}{2})^2 + 1^2 + (\frac{\sqrt{3}}{2})^2 + 2 \times (0 + 0 + 0) = \frac{5}{2}$, 所以

$|\overrightarrow{BD}| = \frac{\sqrt{10}}{2}$. 故选:A.

[V] 对照 hsize=T 长文本 (旧 minipage 断行, 全窄)

[详解] 过点 B, D 分别向 AC

作垂线, 垂足分别为 M, N ,

则可得 $AM = \frac{1}{2}$, $BM = \frac{\sqrt{3}}{2}$,

$CN = \frac{1}{2}$, $DN = \frac{\sqrt{3}}{2}$,

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$\overrightarrow{BD} = \overrightarrow{BM} + \overrightarrow{MN} + \overrightarrow{ND}$, 所以

$|\overrightarrow{BD}|^2 = (\overrightarrow{BM} + \overrightarrow{MN} + \overrightarrow{ND})^2$

$= |\overrightarrow{BM}|^2 + |\overrightarrow{MN}|^2 + |\overrightarrow{ND}|^2 + 2(\overrightarrow{BM} \cdot$

$\overrightarrow{MN} + \overrightarrow{MN} \cdot \overrightarrow{ND} + \overrightarrow{BM} \cdot \overrightarrow{ND})$

$= (\frac{\sqrt{3}}{2})^2 + 1^2 + (\frac{\sqrt{3}}{2})^2 + 2 \times (0 +$

$0 + 0) = \frac{5}{2}$, 所以 $|\overrightarrow{BD}| = \frac{\sqrt{10}}{2}$.

故选:A.